

MIND & SPIRIT

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A.M.D.G.

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*“Now faith is the substance
of things to be hoped for,
the evidence of things that
appear not.”*

HEBREWS 11:1



01 — FAITH

Angels, Demons, and Spiritual Warfare: A Catholic Guide to Spiritual Protection

So let's begin by talking about the pink elephant in the room.

BY SHIRLEY XU · 16 AUGUST 2026 · 7 MIN READ

So let's begin by talking about the pink elephant in the room.

By that, I mean, of the invisible presence of angels and demons.

Ooh, angels, demons, what a loaded topic.

If you were bred as a secular person, that is, not of the Catholic/Christian variety, unlike

me, you may scorn the beliefs in angels and demons as heretic, as superstitious, as occult, as insanity.

But, that may mean you don't believe in God.

Because to believe in God is to believe in the spiritual realm where His Spirit resides and to believe in the spiritual realm is to believe in the existence of angels and demons.

Is it terrifying to you? To believe in their existence?

Don't worry, though, we'll get into the nitty gritty on how exactly to protect yourself against demons (see spiritual warfare).

Catholicism teaches that the world is both visible and invisible. Alongside the physical world which we as living, breathing humans reside in and breathe in and experience every day of our lives, Scripture speaks of angels who serve God, demons who oppose Him and serve Satan, and the spiritual battles that occur every single day, as long as this world is living, in which Christians are called to fight for their Lord and Savior in their faith in Christ.

St. Paul writes:

“For our struggle is not with flesh and blood but with the principalities, with the powers, with the world rulers of this present darkness, with the evil spirits in the heavens.” - Ephesians 6:12

So, a core belief that Catholics have is that God is the Lord, the Savior, the King of Angels.

God is Lord of all creation, the Savior of humanity, and the sovereign King of heaven and earth. All the holy angels belong to Him, worship Him, and serve His will.

In Catholic theology, God created the angels. The angels are spiritual creatures designed to serve God and designed to protect us from the dark forces that terrorize the fallen world (the world we live in).

As Catholics, those baptized in the Catholic Church, you are given guardian angels.

Ooh, guardian angels, myth and fantasy.

Secular media has taught that angels aren't real, demons are fantasy and also not real, and stuff of fiction.

So let me pose this interesting question, that, if you're a real Catholic you might not want to answer because you're afraid of being called a heretic, or if you're not a real Catholic, you don't know the answer to or think it's a stupid question.

Why silver? Why not gold? And what is the difference between a silver crucifix and a gold crucifix?

We'll get back to the answer to this interesting question later.

The Catechism teaches:

“The existence of the spiritual, non-corporeal beings that Sacred Scripture usually calls ‘angels’ is a truth of faith.”
— *Catechism of the Catholic Church, 328*

Throughout Scripture, angels worship God, deliver His messages, protect His people, and participate in His providential plan.

Catholic tradition honors the three Archangels named in Scripture:

St. Michael - Defends people against Satan

St. Gabriel - Announces God's messages

St. Raphael - Companion, Protector, and Instrument of God's Healing.

Catholics believe in guardian angels. Jesus' word in Matthew 18:10 supports the care of angels for human beings.

So, now, what are demons?

Demons are fallen angels. Spiritual beings originally created by God for good who have rejected Him and followed Satan instead.

The Catechism explains:

“The devil and the other demons were indeed created naturally good by God, but they became evil by their own doing.”

So let's pose this interesting question.

What's evil? People? Certain people? Or demons?

The answer?

Let's get back to this shortly.

So let's now talk about spiritual warfare.

Spiritual battles are less dramatic than the movies portray. Spiritual battles can simply mean your mind filled with sinful thoughts given to you by the demons. I mean, ever read *The Screwtape Letters* by CS Lewis? I highly recommend it.

Can demons generate sinful, evil thoughts in your mind?

Sure. Without doubt. Absolutely.

The temptation toward pride, hatred, despair, lust, greed, hatred, dishonesty, envy, resentment, hatred, murder, lust, hatred, lust, evil and abandonment of all things good is the daily battleground in the Christian life.

But don't worry. Do you know the real reason people become baptized as Catholics? I mean, this is pretty controversial, but the difference between Catholicism and Protestantism, is that the Eucharist, Holy Water are real and the Holy Spirit is not a metaphor but Real,

and Catholics become filled with the Holy Spirit when they are Baptized. For real.

So, awesome. If you want to be a good person, just be baptized in the Catholic Church. You'll be freed from the slow, painful death that evil perpetuates in the sinners who aren't baptized in the Catholic Church.

Protestants, I'm sorry, but symbolism isn't going to save you from the Devil.

Jesus, His Blood and Body will.

So now let's talk about the Amor I mean Armor of God.

Ephesians 6 is a pretty powerful passage.

No it's not symbolic.

St. Paul tells you to put on the Armor of God, meaning

- the belt of truth
- the breastplate of righteousness
- readiness for the Gospel of peace
- the shield of faith
- the helmet of salvation
- and the sword of the spirit, which is the Word of God

The real spiritual weapons?

truth, righteousness, faith, salvation, Scripture, prayer, and perseverance

If you stay Christ-centered and live without sin, you will most certainly prevail in spiritual warfare.

But can people really be sinless?

This is why the Sacraments exist.

Why do people go to Confession?

Spiritual warfare is ultimately connected with the struggle against sin. When you sin, you open yourself up to the demons, the darkness of this world. Rather than being preoccupied with the occult and demonic activity, Catholics should go to Confession and clean themselves from sin.

It's really hygiene, in essence, to go to Confession.

Sin leaves a bloody stain on your soul. Demons are attracted to the scent of sin. Go to Confession to clean the sin off of you so demons will leave you alone.

Is this too hard to understand?

Does this sound like blasphemy?

Not your problem.

Confession, repentance, commitment to Christ by partaking in the Eucharist, daily prayer, Rosary, going to Mass on Sundays...

All essential in spiritual warfare, in NOT succumbing to demons.

So this might sound like the occult.

Which is believing in the supernatural.

"Summoning spirits, seances, divination, psychics, astrology, spells, witchcraft" and seeking supernatural power apart from God.

So what's the difference between what I'm saying about demons and the occult?

See, astrology twisted the stars to attribute power away from God to the Greek gods. Greeks invented astrology, the whole synastriy, and horoscopes to diminish God's power of us. That's why the Greek pagans were heretics.

Whether the stars determine our nature during our birth?

That's for you to discern, to discuss, and to decide.

So more about me? Do I believe in the occult? You're looking at someone who absolutely hates everything supernatural and horror related.

Vampires? Not real. Werewolves? Not real? Witches?

What are witches?

Basically people who dabble in the supernatural and worship Satan and use the powers of darkness to unleash demons on you. Basically they can attack your mind with their witchy ways, that's how they get you.

Were they real?

In the Harry Potter way?

Doubt it.

Ever? Like Salem?

Yeah, totally.

So back to sacramentals.

Crucifixes, rosaries, scapulars, medals, holy water.

They all help against the darkness.

Why Silver?

Why not God?

Old school Catholics believe that silver is the most potent in warding against evil and demons.

Gold isn't as potent, demons aren't afraid of gold.

But silver historically has purifying properties.

So yeah, silver > gold.

If you want to blame someone else for the thoughts attacking your mind and preventing you from acing that exam.

Fine.

But believe me, it's the demons.

But hear me out.

God is real. Use your angels and sacramentals. Pray for victory in spiritual warfare. And please be baptized in a Catholic church, please go to Confession.

If you want to fall to the darkness and ruin your life,
fine.

That's your prerogative.

Your decision.

Up to you.

But really?

It's your choice.

Remember, you're not evil.

The demons are.

And whether you are protected from them or prey for them?

That's your choice.

You were made with love by God.

But do you love?

Are you love?

Remember the fruit of the Spirit.

But the fruit of the Spirit is love, joy, peace, forbearance, kindness, goodness, faithfulness, gentleness and self-control. Against such things there is no law. - Galatians 5:22-23

The real advantage of being Catholic and choosing the ways of God isn't to look cool, to be a part of a certain cool culture, or to even get a husband or wife who's Christian.

It's so you can win against evil.

So learn and pause before accusing someone of being a heretic.

Blame the Devil, even.

Peace.



Romanticize

[rō'man(t)ə,sīz] verb

Taking in every simple moment and

02 — FAITH

How To Romanticize Your Life: Finding Meaning in God's Gifts

Learn how to romanticize your life with gratitude, mindfulness, and faith. Discover simple ways to find beauty, joy, and purpose in everyday moments.

BY SHIRLEY XU · 16 JULY 2026 · 8 MIN READ

"This is the day that the Lord has made; let us rejoice and be glad in it." — Psalm 118:24

You have choices in life. Would you rather be stuck in your head, bemoaning all of the problems that pop up in daily life? Or would you rather start living

your life, being the main character and celebrating each day for the gift that it is?

Life is a gift. Sometimes we're so caught up in just surviving that we forget the fortunate circumstances we're in. Modernity has advanced tremendously over the past few decades, giving us opportunities that previous generations could only dream of. We have the inter-

net, incredible technology, access to education, and opportunities to build careers, businesses, and relationships across the globe.

Yet we often focus on what we don't have instead of what we do. That's the scarcity mindset—a trap that steals our joy. Scripture reminds us to shift our perspective:

"Give thanks in all circumstances; for this is the will of God in Christ Jesus for you."
— 1 Thessalonians 5:18

Gratitude doesn't mean pretending life is perfect. It means recognizing God's blessings even in imperfect seasons.

Celebrate your life. How? By living it. By fully embracing it. By narrating it and by documenting it.

Stop Waiting for Life to Begin

The truth is, most of us spend far too much time waiting.

We wait until we lose the weight.

We wait until we get the promotion.

We wait until we move into a nicer apartment.

We wait until we have more money.

We wait until someone falls in love with us.

We wait until life finally begins.

But life isn't waiting somewhere in the future.

It's happening right now.

Every ordinary Tuesday afternoon. Every commute home. Every cup of coffee in the morning. Every rainy walk. Every conversation with a friend. Every quiet evening spent

reading a book. These moments seem insignificant when they happen, but together they become your life.

Jesus Himself reminds us not to become consumed by tomorrow:

"Therefore do not worry about tomorrow, for tomorrow will worry about itself. Each day has enough trouble of its own." —
Matthew 6:34

God invites us to be fully present today.

When people look back on their happiest memories, they rarely remember the expensive things they bought. They remember laughing until they cried over dinner with friends. They remember family vacations, sunsets on the beach, road trips with the windows down, baking cookies with their grandmother, or studying late into the night with classmates who eventually became lifelong friends.

Life is built from moments, not milestones.

Become the Main Character

One of the most popular trends on social media is the idea of becoming "the main character." While the phrase has become somewhat cliché, the message behind it is surprisingly meaningful.

Being the main character doesn't mean believing the world revolves around you. It means recognizing that this is your story.

More importantly, it is recognizing that God is the Author.

"For we are God's handiwork, created in Christ Jesus to do good works, which God prepared in advance for us to do." — Ephesians 2:10

You are the one writing each chapter, but God is faithfully guiding the story.

You decide whether your mornings begin with gratitude or anxiety. You decide whether your evenings are spent endlessly scrolling through someone else's highlight reel or creating memories of your own.

Imagine watching a movie where the protagonist spends every scene comparing themselves to everyone else or worrying about things they cannot control. It wouldn't be a very interesting story.

The most compelling characters are curious. They take risks. They learn new skills. They travel. They laugh. They cry. They make mistakes. Most importantly, they continue moving forward.

You deserve to be that person in your own life.

Romanticize the Ordinary

One of the easiest ways to cultivate happiness is to make ordinary moments feel special.

You don't need a five-star vacation to experience beauty.

Light a candle while you're reading.

Buy yourself flowers just because.

Listen to music while cooking dinner.

Watch the sunrise before work.

Take yourself on a solo coffee date.

Wear the outfit you've been saving for a "special occasion."

Who says today isn't special?

We often reserve joy for birthdays, holidays, anniversaries, and vacations. But there are 365 days in a year, and every one of them is an opportunity to experience God's goodness.

The ordinary becomes extraordinary when we choose to pay attention.

As King Solomon wrote:

"He has made everything beautiful in its time." — Ecclesiastes 3:11

Beauty isn't something we create from nothing—it is something we learn to notice.

Start Documenting Your Life

Record your life. There's a reason why social media is so popular. It's a form of gratitude to take pictures and videos of our lives and show them to the world. You are the main character of your life. And so are others for their lives.

Document your life every day.

Take pictures.

Write in a journal.

Film short videos.

Keep a scrapbook.

Write down funny conversations.

Record answered prayers.

Write about lessons God is teaching you.

Each day is a gift from God, something to be celebrated and to unwrap. Let us fully im-

merse ourselves in the meaning and beauty of each day.

Throughout the Old Testament, God repeatedly instructed His people to remember His faithfulness. They built memorial stones, celebrated feasts, and retold stories so they would never forget what He had done.

Our journals and photographs become modern memorials of God's faithfulness throughout our lives.

Gratitude Changes Everything

The fastest way to transform your perspective on your life is through gratitude.

Our brains are wired to notice problems around us. It's an evolutionary survival mechanism. We are trained to find problems and to figure out ways to solve them, to defeat them, to surmount them. While this helped our ancestors survive in the wilderness, this also means we naturally overlook the blessings surrounding us.

Gratitude interrupts the default mechanism we have to notice and muse on problems.

Instead of focusing on what is going wrong in our lives or on what is missing, let's begin noticing everything that's going right, that's already in our lives, that's been given to us.

Don't take what you have for granted. Celebrate them. Celebrate your family and friends. Celebrate your possessions, even. Celebrate the food on your table. Celebrate your job, your skills, your education.

James reminds us:

"Every good and perfect gift is from above, coming down from the Father of the heavenly lights." — James 1:17

Create More Than You Consume

There's so much information out there; news, social media, web content, mobile apps, books, TV shows.

Our generation consumes information at an unprecedented rate.

We scroll through hundreds of videos every day. We binge television shows. We read comments, headlines, and endless streams of content.

But how much are we creating?

Write something.

Paint something.

Bake something.

Build something.

Learn an instrument.

Start a garden.

Take photographs.

God is the ultimate Creator, and because we are made in His image, we were created to create.

Whether you're writing a novel, designing a website, baking bread, or planting flowers, you're reflecting the creativity of the One who made you.

"Whatever you do, work heartily, as for the Lord and not for men." — Colossians 3:23

Even ordinary work can become an act of worship.

Stop Waiting for Permission

We've lived a large part of life waiting for permission. By our parents, by our bosses, by authorities, by people who are rich and famous, by people who we don't even know.

But you don't need permission to enjoy your life.

You don't need to earn happiness.

You don't need to wait until you've achieved every goal on your bucket list before allowing yourself to smile and to be happy.

Celebrate your small wins.

Celebrate making your bed.

Celebrate finishing a workout.

Celebrate reading another chapter of a book.

Celebrate making it through a difficult week.

These moments matter, they form the fabric of your growth and of your story, and they are evidence of God's imprint on your life.

Jesus said:

"I came that they may have life and have it abundantly." — John 10:10

An abundant life isn't measured by net worth or possessions or fame. It is a life measured by God's grace evident in your days.

Your Story Is Already Beautiful

Sometime we look at other people's lives and think, *my life is boring compared to theirs, I'm not as successful or as beautiful or as*

rich or as famous, I'm not living a life as perfect as theirs.

But you're forgetting one important thing.

Your life doesn't have to look like someone else's to be meaningful.

No one's life is exactly like someone else's. The life that God is writing for you is unique to you and solely for you. And isn't that something to be glad about it, to appreciate and to be grateful for, and to even be ecstatic about?

You are uniquely yours, your life is uniquely yours, and isn't it such a beautiful truth that God has woven a beautiful story that looks like no one else's but contains your fingerprint and can't be replicated by anyone else?

The truth is, you don't need to be like someone else or to imitate someone else's life or success in order to be successful or have a successful life.

Celebrate others' lives for their successes. Their successes are uniquely theirs.

But your successes are uniquely yours as well.

And as you do, remember God's promise:

"For I know the plans I have for you," declares the Lord, "plans for welfare and not for evil, to give you a future and a hope." — Jeremiah 29:11

One day you will look back on the story of your life and realize,

You've lived a great life.

Full of beauty and pain and meaning and success and accomplishment and impact and extraordinary heart,

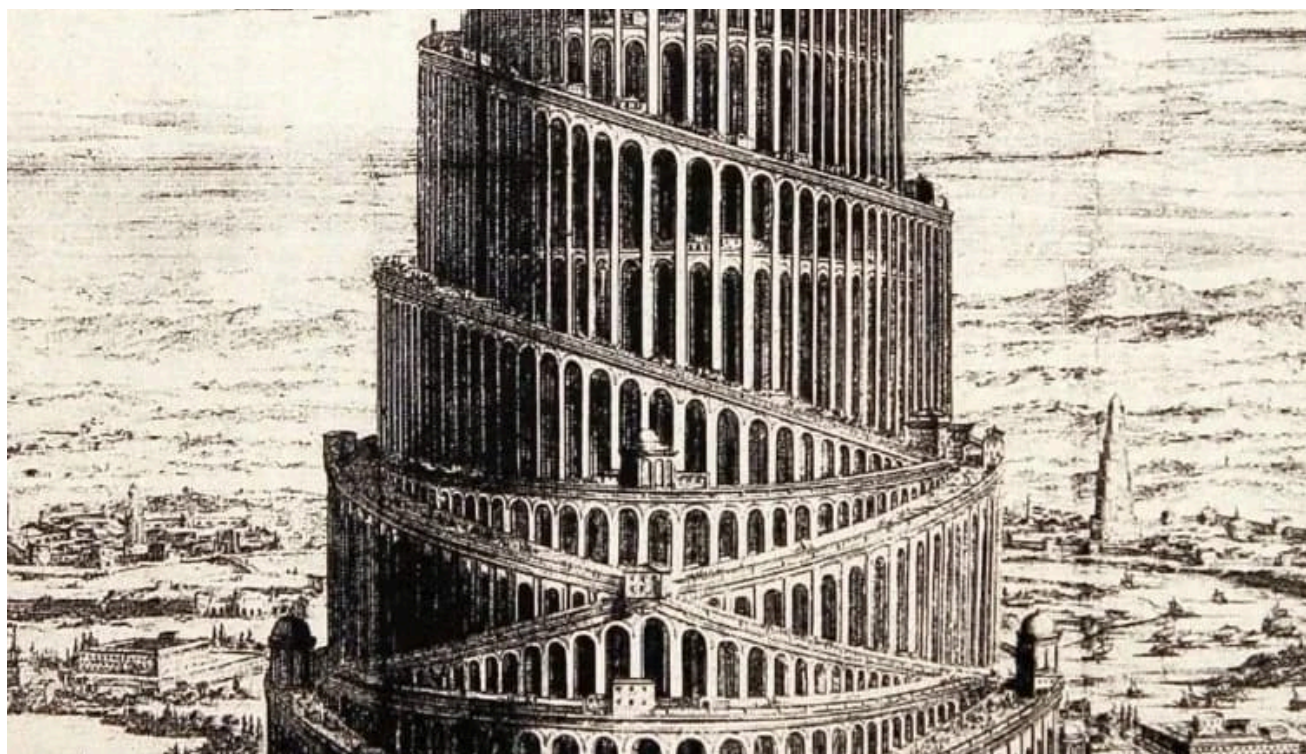
your life wasn't ordinary,

It was a life lovingly written by God, filled with grace in the ordinary, beauty in the everyday, and countless gifts waiting to be noticed.

So romanticize your life—not because every day is perfect, but because every day is a gift from God who gave it.

"This is the day that the Lord has made; let us rejoice and be glad in it." — Psalm 118:24





03 — FAITH

Where is the human heart? On Magnifica Humanitas, Pope Leo XIV's Encyclical on AI and Human Dignity

In one's writing, there exists an expression of the human heart. Where truth dwells, truth of the person, truth of the mind, truth of the spirit.

BY SHIRLEY XU · 2 JUNE 2026 · 5 MIN READ

Where your treasure is, there your heart will be also.

In one's writing, there exists an expression of the human heart. Where truth

dwells, truth of the person, truth of the mind, truth of the spirit.

This truth I came across in my own writing. Specifically, for this blog.

I've been accused of using AI to write my blog posts and creative writing. My own creations were dismissed as lies and AI-generated. Being sensitive to the castigation of my enemies, I thought — it no longer mattered whether I wrote it or not. If integrity is no longer valued, if one's integrity is not acknowledged, if one is not acknowledged but AI is... how about we replace ourselves with AI?

Society has been ridden with lies since the beginning of the fall. In my philosophy class at Northwestern where I attended college, I once quoted Nietzsche in an essay —

“All things are subject to interpretation. Whichever interpretation prevails at a given time is a function of power and not truth.”

I said that truth claims were merely power plays. I repeated the line in a Dominican Discourse gathered to discuss the murder of Charles Kirk, and the priests laughed, smirked, sneered at me, not understanding what I meant. (Perhaps, at the sexual undertones? Perhaps, at my apparent ignorance?)

To dominate is to suppress another in a display of power, of overwhelming another's perspective on truth in an attempt to convince another of their truth — a debate of sorts. In a court of law, the victor is the one who can convince the judge, the jury, that their perspective, their argument is the truth.

And it doesn't matter if it really is the truth.

On a stage, we meet — again? To dance, with the Devil?

Does it ever frustrate you when someone plays Devil's Advocate with you? But maybe

they have to, to preserve their own mind, their own sanity, even. A tug of war — of sorts — between one and one's opponent.

Is it — victory? To dominate another? In a power struggle? To claim that one's truth is the REAL, OBJECTIVE, ABSOLUTE TRUTH.

What is the most important? To win? Or to know, to say, to be truth?

Because, it's a fundamental truth, that people lie.

To gain control of an argument, to gain control over someone else, to be the one holding the reins of power.

And now, with the evolution of AI (finally), the game has changed. An instrument to be used either for good or for evil.

Pope Leo, in his Magnifica Humanitas, speaks of the choice the humanity must make — build a new Tower of Babel or rebuild Jerusalem. Two biblical images frame the argument:

Tower of Babel:

- symbolizes pride, centralized power, uniformity, and attempts to dominate without moral grounding.
- Applied to AI, it represents technologies driven by control, profit, and dehumanization.

Rebuilding Jerusalem (Nehemiah):

- symbolizes cooperation, community, shared responsibility, and rebuilding society together.
- Applied to AI, it means using technology for solidarity and the common good.

I came across many AI-generated videos while I was browsing on Tik-Tok. Content creators used AI for profit, to masquerade AI-created videos as their own. Doesn't this compromise the integrity of the creator?

Think of it, of course, we wish to profit, to save time, for "time is greater than space" as Pope Leo quotes Pope Francis, but is it really necessary to lie to the public? Can we trust the creator if their content is all built on lies and a dehumanization? Because, where is their heart, where is their spirit, where is their truth in what they have made?

As an experiment, I used AI to write my blog post drafts. And compared them to the blog posts that were really written by me.

And, evaluating them, I've noticed — a difference, in tone, in warmth, in flow and impression, in comfort and sanity. And although the AI created blog post might be superior in knowledge depth and annotation, my real writing actually had spiritual depth. It marked my soul, a watermark of my spirit.

Where is the human heart in AI?

Pope Leo says:

I ask everyone to abandon the construction of yet another Tower of Babel and to join forces in building up the common good, so that humanity will never lose its beauty, and the world once again will come to recognize the human heart as the place where God desires to dwell.

Mind. Heart. Spirit. Body.

What the Triune Lord created us to be, in parts, in whole.

And isn't it sacrilege to replace ourselves with, what is possibly, the lies of the Devil?

I thought of a scary thing, while browsing Tiktok...

In creating a new Tower of Babel, people lie to the public and commit a type of fraud. Fake news is a very real thing, and isn't it scary to think that you could be brainwashed into believing propaganda from other persons, institutions, governments, cults?

Isn't this what Hitler did, essentially? To brainwash an entire semi-Satanic cult of Nazi-led followers into committing mass genocide/torture/heinous annihilation of a religion that essentially bred and borne Jesus, the Savior who will one day lead us to...

I mean, isn't that the sign of a narcissist?

Sometimes...

we want what we want when we want it. (a priest said this)

What if AI could be used like this? By our enemies? What if AI could be used to create androids, robots, FAKE HUMANS that look like us, talk like us, write like us, reproduce not unlike us but asexually?

What if fake people walk among us? you mean?

Good question.

But here's the good news.

Sometimes, lies are a good thing. They can bring us closer to God.

Sometimes, disaster, chaos, sin bring us closer to God.

In trying to decipher the lies/half-truths of others, we rely more heavily on Jesus, the Holy Spirit, and the Lord to TELL us what the truth really is.

And in walking the Way, the Truth, the Life, we can really reach the truth.

What the Tower of Babel was really striving for.

Sometimes we make mistakes.

Sometimes we fail.

Sometimes we don't know what the answers are.

Sometimes we get lost.

Sometimes we try again.

And by training ourselves, to be more like Jesus, to think more like Jesus, to write more

like Jesus, we can reach what we wanted all along.

Goodness. Beauty. Perfection. Infinity.

A fullness, a completion, a wholeness.

And yeah,

we can never really be Jesus.

No one can replace him.

Not even Jonathan Roumie.

AI can't replace us.

But, we can use AI to change us.

To perfect us, to fill in the gaps of what we don't know, so we do know.

And isn't this what Jesus, our Teacher, wanted for us, his disciples, all along?

| *Come and see.*





04 — MIND

The Hijacked Mind: How Social Media Rewires the Brain — and How to Reclaim Your Attention

Why scrolling makes your mind wander: the neuroscience of the Default Mode Network, rumination, and strategies to regain deep focus.

BY SHIRLEY XU · 15 MARCH 2026 · 7 MIN READ

Modern social media platforms are often framed as tools of distraction. Yet distraction is only the surface symptom of a deeper neurological shift. Increasingly, neuroscientists argue that the architecture of social media interacts directly with fundamental brain networks responsible

for attention, self-reflection, and emotional processing.

To understand why scrolling through a feed can lead to spiraling thoughts, comparison, anxiety, and persistent mental noise, we must examine a system known as the Default Mode Network.

This network governs much of our inner mental life. It shapes how we think about ourselves, imagine the future, interpret social relationships, and construct narratives about our lives. When balanced with the brain's executive control networks, the DMN supports creativity, planning, and introspection. But when it becomes chronically dominant, it can trap the mind in loops of rumination and wandering thought.

Social media, as it turns out, is uniquely designed to keep this system active.

The Brain at Rest: The Default Mode Network

The Default Mode Network was first identified in the early 2000s when neuroscientists noticed that certain brain regions consistently became more active when subjects were not performing an explicit task.

Rather than going quiet during rest, the brain appeared to shift into a different mode of activity — one devoted to internal cognition.

The DMN consists of several interconnected structures:

- Medial Prefrontal Cortex — central to self-referential thought and evaluation
- Posterior Cingulate Cortex — associated with autobiographical memory and conscious awareness
- Angular Gyrus — integrates social and conceptual information
- Hippocampus — involved in memory reconstruction and imagining the future

Together these regions generate what psychologists sometimes call the narrative self.

When the DMN is active, the brain engages in processes such as:

- autobiographical memory retrieval
- mental time travel
- imagining future scenarios
- evaluating social relationships
- constructing identity narratives

This network is responsible for the mind's tendency to drift inward — to replay conversations, imagine possibilities, or evaluate personal status.

In evolutionary terms, this system likely helped humans simulate social situations and anticipate threats within complex communities.

But in modern digital environments, the DMN is constantly stimulated in ways it was never designed to handle.

The Brain's Other System: Executive Control

Balancing the Default Mode Network is a set of networks responsible for goal-directed cognition.

Central to this system is the Dorsolateral Prefrontal Cortex, a region that plays a crucial role in executive control.

This area supports:

- sustained attention
- working memory
- decision-making
- impulse control
- suppression of irrelevant thoughts

When you study for an exam, solve a math problem, write code, or read a complex book, the executive control network becomes dominant.

During these moments, the brain suppresses activity in the Default Mode Network.

In other words, focus requires quieting the inner narrative.

Deep concentration and rumination cannot occupy the brain at full strength simultaneously.

Healthy cognition depends on the dynamic balance between these two systems: the outward-focused executive network and the inward-focused Default Mode Network.

Social media disrupts this balance.

The Perfect Stimulus for the DMN

Social media platforms such as Instagram, TikTok, and X (Twitter) operate through short bursts of emotionally charged information presented in an infinite sequence.

Each post requires minimal cognitive effort to process.

Unlike reading a book or solving a problem, scrolling does not strongly engage executive networks.

Instead, posts function as social stimuli that feed the brain's narrative machinery.

A photograph of a vacation activates comparison.

A post announcing a promotion activates evaluation.

A relationship announcement activates speculation.

These cues stimulate self-referential processing in the medial prefrontal cortex, pulling the brain into internal evaluation:

Where do I stand?

Am I behind?

What does this say about my life?

Because the task demands are low, the executive system never fully activates to suppress these thoughts.

The result is a brain simultaneously consuming external content while generating internal narratives.

The Neurochemistry of the Scroll

Social media also interacts with the brain's reward system.

Notifications, likes, and comments trigger the release of Dopamine, a neurotransmitter that signals motivational significance and reward prediction.

Importantly, these rewards arrive intermittently.

Sometimes a post receives many likes.

Sometimes it receives none.

Sometimes a message arrives immediately.

Sometimes hours later.

This unpredictability creates a variable reward schedule — the same mechanism that makes gambling machines addictive.

Variable rewards strengthen neural circuits involved in anticipation and habit formation.

Over time, the brain begins generating persistent predictive loops:

Did someone respond?

Did my post perform well?

Should I check again?

Even when the phone is not present, these loops can continue internally, sustaining activity within the Default Mode Network.

Emotion, Attention, and the Amygdala

Social media algorithms tend to amplify emotionally charged content because it generates engagement.

Posts that provoke outrage, envy, excitement, or anxiety travel further through networks.

These emotional cues activate the Amygdala, a structure central to detecting emotionally significant stimuli.

When the amygdala signals importance, attention systems are recruited.

But instead of leading to sustained focus, social media feeds rapidly introduce the next stimulus.

The brain becomes trapped in cycles of:

emotional activation → narrative interpretation → comparison → next stimulus.

Emotionally salient stimuli therefore become integrated into self-referential narratives generated by the DMN.

This interaction helps explain why social media can amplify rumination and anxiety.

Fragmented Attention and Cognitive Switching

Another neurological effect of social media is the fragmentation of attention.

Rapid switching between posts, videos, messages, and notifications trains the brain to operate in a continuous partial attention state.

Each switch requires the executive system to reorient attention.

Over time, frequent switching weakens the stability of sustained attention networks.

The brain becomes accustomed to novelty and micro-stimulation rather than prolonged focus.

Tasks that require deep engagement — studying, writing, programming, learning — begin to feel cognitively demanding by comparison.

Not because they have become harder, but because the neural circuits responsible for sustained attention are undertrained.

The Consequence: A Noisy Mind

When the Default Mode Network remains chronically active, the mind becomes filled with internal chatter.

Psychologists associate persistent DMN activation with:

- rumination
- anxiety
- depressive thinking
- intrusive memories
- wandering attention

This does not mean the DMN is inherently harmful. On the contrary, it is essential for

creativity, empathy, and identity formation.

The problem arises when it dominates cognition without sufficient regulation from executive networks.

Modern digital environments can tilt this balance.

Reclaiming Executive Control

Fortunately, the brain retains a remarkable capacity for change. Neural circuits strengthen with repeated use through a process known as experience-dependent plasticity.

Practices that repeatedly recruit executive attention can rebalance the brain's networks.

Several interventions have shown strong effects in neuroscience research.

Focused Attention Meditation

Meditation trains the brain to detect when the mind has wandered and return attention to a chosen object.

Each time attention is redirected, the executive network — particularly the dorsolateral prefrontal cortex — becomes engaged.

Over weeks of practice, neuroimaging studies show reduced activity in the Default Mode Network during rest.

Even brief daily sessions can improve attentional stability.

Deep Work

Extended periods of uninterrupted concentration strongly activate executive networks.

Activities such as writing, studying, coding, and design require the brain to maintain goal representations over time.

When practiced regularly, these tasks strengthen neural pathways supporting sustained attention.

Researchers studying high performers often observe work cycles lasting 60 to 90 minutes, followed by recovery periods.

Aerobic Exercise

Physical movement has powerful cognitive effects.

Aerobic exercise increases cerebral blood flow and promotes the release of neurotrophic factors that support synaptic plasticity.

Regular exercise is associated with improved executive function and reduced rumination.

Even moderate activities like brisk walking can reduce DMN overactivity.

Long-Form Reading

Reading books or long essays requires maintaining a coherent mental representation across many pages.

Unlike social media posts, which deliver fragmented information, long-form reading encourages sustained narrative engagement.

This process recruits both executive networks and memory systems.

Over time, reading strengthens attentional endurance.

Creative Flow Activities

Activities that combine challenge with skill can induce flow states, a condition in which attention becomes fully absorbed in the task.

Examples include:

- drawing
- playing music
- programming
- cooking
- writing

During flow states, DMN activity decreases while task networks become dominant.

Many people experience these activities as deeply restorative precisely because they quiet internal rumination.

Protecting Attention in the Digital Age

The modern attention economy is designed to capture cognitive resources.

Social media platforms are not neutral tools; they are environments engineered to maxi-

mize engagement.

Understanding the neuroscience behind these systems allows us to interact with them more deliberately.

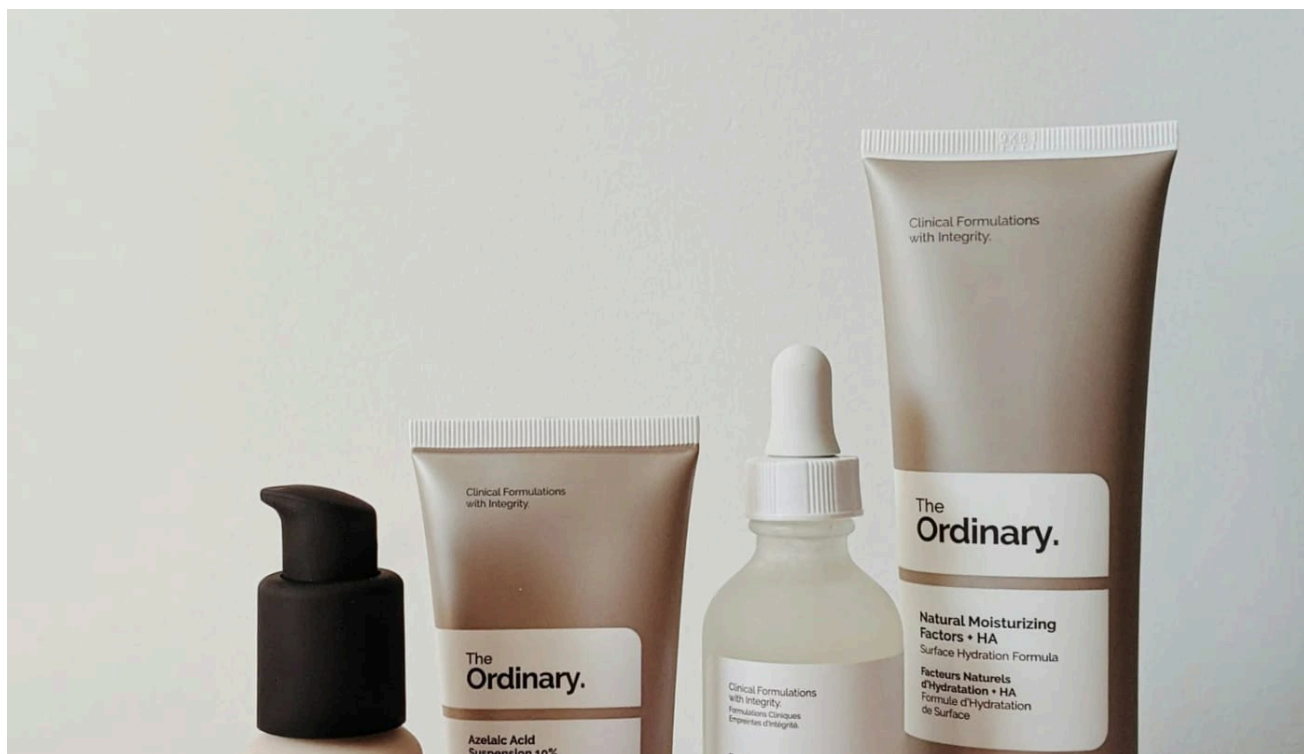
Attention is not merely a psychological preference — it is a neurological resource.

The patterns we repeat shape the architecture of the brain itself.

When we cultivate habits that strengthen executive control, we gradually reclaim the capacity for sustained focus, deeper thinking, and a quieter mind.

In a world of endless scrolling, the ability to direct attention may become one of the most valuable forms of freedom.

†



05 — WELLNESS

My Skincare Routine - The Science and Tea Behind Skincare Products

I got into skincare a couple of years ago, having developed an interest in Korean skincare. One of my friends had a blog where she reviewed and rated Korean skincare products, and that got me hooked. Before I knew it, I was obsessively googling and looking up skincare products on YesStyle and Amazon, not to mention Sephora and Ulta Beauty. I was really interested in the science behind skincare, having double majored in biology and chemistry in college. I'd like to explain the science behind the various steps of skincare routines and share some skincare products I've been loving as well.

BY SHIRLEY XU · 26 SEPTEMBER 2025 · 7 MIN READ

I got into skincare a couple of years ago, having developed an interest in Korean skincare. One of my friends had a blog where she reviewed and rated Korean skincare products, and that got me hooked. Before I knew it, I was obsessively googling and looking up skincare products on YesStyle and Amazon, not to mention Sephora and Ulta Beauty. I was really interested in the science behind skincare, having double majored in biology and chemistry in college. I'd like to explain the science behind the various steps of skincare routines and share some skincare products I've been loving as well.

Cleanser

The very first step to any good skincare routine is washing your face with cleanser. The outermost layer of your skin is the stratum corneum, which is made up of dead skin cells and lipids (fats). The role of the stratum corneum is to keep water in and prevent irritants, bacteria, and pollutants from entering the skin. Cleanser is used to remove dirt, oil, and pollutants from the skin without damaging the skin barrier. Cleanser contains surfactants that bind to oil and dirt and washes them away with water. Use gentle, pH balanced cleanser as to not disrupt the natural skin barrier and strip it away of its natural lipids, which will dry and irritate the skin. Cleansers can also contain hyaluronic acid, which keeps water in the skin, and ceramides, which are lipids that make up 50% of the skin that repair the skin, contain moisture, and prevent water loss.

Some of my favorite cleansers include:

CeraVe Facial Cleanser

Daily face wash with hyaluronic acid, ceramides, and glycerin to help hydrate skin without stripping moisture. Removes face makeup, dirt, and excess oil, provides 24-hour hydration and leaves a moisturized, non-greasy feel.

Cosrx Low pH Good Morning Gel Cleanser

Key Ingredients: pH balanced, tea-tree oil, and BHA to soothe and strengthen for luminous and clear skin.

La Roche-Posay Toleriane Purifying Foaming Facial Cleanser

Gentle, non-drying cleanser with Thermal Water, Ceramide-3, Niacinamide & Glycerin. Supports skin barrier.

Toner

Toner is often used after cleanser to hydrate the skin and return the skin to its baseline pH. Types of toner include hydrating toners, which are most common, exfoliating toners, soothing/barrier-supporting toner, and astringent toners.

Hydrating toners contain humectants like glycerin, hyaluronic acid, or aloe that pull water into the skin and help prep the skin for serums and creams.

Exfoliating toners contain low concentrations of AHAs (glycolic, lactic) or BHAs (salicylic acid). They provide mild, daily exfoliation and are great for brightening, acne control, and improving texture.

Soothing/barrier-supporting toners contain calming agents like niacinamide, chamomile, centella asiatica, and panthenol. They reduce

redness, irritation, and strengthen the skin barrier.

Lastly, astringent toners are old school alcohol-based toners made to shrink pores and remove oil. But science shows they can strip the barrier and worsen oil production. Avoid strong alcohol-based toners unless treating very oily skin short-term.

The best toners for dry/sensitive skin are hydrating and soothing (hyaluronic acid, glycerin, panthenol, ceramides). For oily/acne-prone skin, exfoliating (salicylic acid and glycolic acid) toners work best. And for redness, use niacinamide, centella asiatica, and chamomile to calm the skin.

Toner is not a strict requirement for your skincare routine, but it helps to soothe the skin and prep the skin for serums and moisturizers.

Here are some recommended toners:

The Ordinary Glycolic Acid 7% Exfoliating Toner

THAYERS Alcohol-Free, Hydrating Cucumber Witch Hazel Facial Toner with Aloe Vera Formula

Exfoliants

Exfoliants are skincare ingredients or tools that remove dead skin cells from the surface of the skin. Your skin naturally sheds cells in a process called desquamation, but with age, sun damage, or acne, this process slows so the skin looks dull, rough, or clogged. Exfoliants speed up cell turnover, revealing smoother, brighter skin.

Types of exfoliants include physical/mechanical exfoliants and chemical exfoliants. Physical exfoliants use small particles or tools to manually scrub off dead skin. Scrubs with sugar, salt, rice powder, jojoba beads, and brushes, sponges have the pros of immediate smoothness and they are easy to use. The cons are they can be too harsh and tear the skin.

Chemical exfoliants use AHA (alpha hydroxy acids) which are water soluble and BHA (beta hydroxy acids), which are oil-soluble, to work at the molecular level to loosen bonds between dead cells so they shed more easily. AHAs exfoliate the skin surface and are best for dry, dull, or sun-damaged skin. Examples include glycolic acid, lactic acid, and mandelic acid, which brighten skin, reduce fine lines, and improve texture. BHAs penetrate pores and dissolve oil and dead cells. They are best for oily and acne-prone skin. Salicylic acid helps unclog pores and reduce blackheads to fight acne. PHAs include gluconolactone and lactobionic acid to hydrate the skin while exfoliating and are good for sensitive skin.

Exfoliants improve texture by smoothing rough patches, brightens tone by removing dull, dead layers, unclog pores which helps with acne, and stimulate collagen production. You don't need to exfoliate every day, optimal frequency is 1-3 times a week.

Paula's Choice SKIN PERFECTING 2% BHA Liquid Exfoliant

Aveeno Positively Radiant Skin Brightening Exfoliating Daily Facial Scrub

Serums

Serums are concentrated skincare formulas designed to deliver active ingredients deep into the skin. They are lightweight, fast-absorbing liquids or gels. They treat acne, wrinkles, dark spots, and dehydration. They have smaller molecules and fewer occlusive agents than creams and penetrate more effectively. They contain ingredients such as niacinamide serum, vitamin C serum, or hyaluronic acid serum.

Hydrating serums are hyaluronic acid, glycerin, and panthenol. Humectants draw water into the skin and plump fine lines and boost elasticity. Brightening serums are vitamin C, niacinamide, licorice root, alpha arbutin. Antioxidants neutralize free radicals and regulate melanin production. Anti-aging serums are retinol, peptides, growth factors, and they increase collagen/elastin production, improve cell turnover, reduce wrinkles.

Acne/exfoliating serums are salicylic acid, glycolic acid, lactic acid, azelaic acid and unclog pores, normalize shedding of skin cells, and reduce inflammation. Barrier repair/soothing serums are ceramides, niacinamide, centella asiatica, madecassoside which strengthen the lipid barrier, reduce irritation, and calm redness.

Apply serum after cleansing/toner before moisturizer. Don't mix active ingredients. Results usually take 6-12 weeks to show since serums target cellular processes.

Top reviewed serums:

COSRX Snail Mucin Repairing Serum

The Ordinary Niacinamide 10% + Zinc 1%, Smoothing Serum

The Ordinary Hyaluronic Acid 2% + B5 (with Ceramides), Multi-Depth Hydration Serum

Sunscreen

Sunscreen is a topical product that protects your skin from UV radiation, which causes DNA damage (which can lead to skin cancer), breakdown of collagen/elastin (which leads to wrinkles and sagging), pigment changes (which lead to dark spots and uneven tone), and inflammation which causes redness and sunburn.

Sunscreen contain mineral filters, chemical filters, or a hybrid. Mineral filters include ingredients like zinc oxide and titanium dioxide which sit on top of the skin and reflect/scatter UV rays. They are good for sensitive skin and are broad-spectrum. They can leave a white cast.

Chemical filters include ingredients like avobenzone, octocrylene, octinoxate, oxybenzone, Tinosorb, and Mexoryl. They absorb UV rays and convert them to heat. They are lightweight and have no white cast. Some can irritate sensitive skin.

Hybrid sunscreens combine both mineral and chemical filters for better coverage and textures.

SPF (Sun Protection Factor) measures the protection against UVB rays. SPF 30 blocks 97% of UVB and SPF 50 blocks 98-99%.

Sunscreen prevents collagen breakdown, so fewer wrinkles, and protects melanocytes to reduce hyperpigmentation. They shield DNA to lower risk of melanoma, basal cell carcinoma, and squamous cell carcinoma.

Apply sunscreen 15-20 minutes before sun exposure for chemical filters while mineral filters work immediately. Reapply every 2 hours. UVA rays penetrate clouds and window so apply every day, year-round.

Sunscreen is the #1 proven skincare product to prevent skin cancer, photoaging, and pigmentation. Daily use is important for long-term healthy skin.

Top Sunscreens:

La Roche-Posay Anthelios Melt-In Milk Sunscreen SPF 60

EltaMD UV Clear Face Sunscreen SPF 46, Oil-Free with Zinc Oxide

Moisturizers

A moisturizer is a topical product designed to increase and maintain skin hydration and support the skin barrier. They trap moisture, smooth surface, and repair lipids.

Moisturizers combine three main classes of ingredients — humectants, emollients, and occlusives.

Humectants pull water from the air or deeper skin layers into the top layer (stratum corneum). Examples include hyaluronic acid, glycerin, panthenol, and urea. They are best for plumping and hydrating dry/dehydrated skin.

Emollients are skin softeners that fill in tiny gaps between dead skin cells, making skin feel smoother and more flexible. Examples include squalane, fatty acids, shea butter, and cholesterol. They improve texture and flexibility of rough and dry skin.

Occlusives are moisture lockers that form a protective film on top of skin to prevent water loss. Examples include petrolatum (vaseline) and dimeticone (silicone) and beeswax. They are best for extremely dry, cracked, or damaged skin.

The skin barrier is made of lipids (ceramides, cholesterol, fatty acids). Moisturizers replace or reinforce these. They prevent flaking, roughness, and tightness. Hydrated skin looks plumper and fine lines are less visible. Occlusive moisturizers reduce irritation from active ingredients like retinoids or acids.

For dry skin, pick moisturizers that are creams with ceramides, shea butter, and petrolatum. For oily skin, use lightweight gels with humectants (hyaluronic acid and glycerin). For sensitive skin, use barrier-repair formulas (niacinamide, ceramides, and colloidal oatmeal). For aging skin, use richer creams with peptides, cholesterol, and hyaluronic acid.

Moisturizers work on a cellular and structural level to reinforce the skin barrier, balance water levels, and protect against environmental stress.

La Roche-Posay Toleriane Double Repair Face Moisturizer

CeraVe Daily Moisturizing Lotion for Dry Skin, Body Lotion & Face Moisturizer with Hyaluronic Acid and Ceramides

Hope this article was informative and inspires you to start your skincare routine — today! Skincare isn't limited to only women, men should at least apply cleanser, sunscreen, and moisturizer at the bare minimum to protect their skin. Begin today and you will see huge

improvements in the quality and look of your skin!





06 — REFLECTIONS

An Introduction

I started this blog back in 2024 primarily to share my creative writing. Later on, I decided to “preach” and write about Christianity.

BY SHIRLEY XU · 20 SEPTEMBER 2026 · 3 MIN READ

Hi guys. This is Shirley. The “author” of this blog.

I started this blog back in 2024 primarily to share my creative writing. Later on, I decided to “preach” and write about Christianity.

Am I a convert? Am I Protestant? Am I Catholic?

Let’s begin with the telling of the truth.

My mom’s half Chinese (Shanghainese) and half British. That makes me $\frac{3}{4}$ Chinese (Shanghainese) and $\frac{1}{4}$ British. Some people think I’m mixed, some people think I’m Korean, some people think I had plastic surgery.

My mom’s dad was British. He’s the reason my mom’s Catholic.

My dad isn’t my biological dad but is “related” to my biological dad. They come from a

“wealthy” family in China, related to “nobility.” They’re originally Buddhist but “converted” to Catholicism.

My mom did surgery in Shanghai. Then she came here. And had to redo medical school. At Harvard. She was chief of cardiothoracic surgery at Weill Cornell (NYC). Later, she became chief at a prominent hospital in London, and she moved us back.

Then she found a job as chief of surgery in a hospital in Flushing and moved us back.

My dad worked on Wall Street as a programmer and project manager. He went to Harvard.

My other dad went to Jiaotong University and Oxford.

I have/had two dads.

I was born in London. I have British citizenship. I was born on April 24 with a twin brother. My parents gave him up for adoption for some reason, but I found him again later in life. Specifically at Weill Cornell because we both had cancer at age 11. Leukemia.

I was baptized as a baby in the Catholic Church. I moved to Shanghai. My grandparents and my other dad raised me. My other dad put me in a Buddhist monastery to teach me discipline and focus (and martial arts) for a “hot minute”.

Then I went back to London. Then to NYC.

Fast forward.

I grew up Catholic. My mom put me in Catholic school and then public school and then Catholic school and then back to public school (whenever I misbehaved).

The nuns were pretty tough on me.

So I ran away from home and joined a “convent.”

At the convent, the nuns were... not that great of people.

Luckily, my mom found me and forced me to pursue surgery as a career. Even though I actually wanted to do art. Or writing. Or plastic surgery.

But ok, cardiothoracic surgery it is.

I went to Northwestern. Got into the HPME program. Basically a dual BA/MD program. I received both degrees. I had a 4.0 GPA. I majored in bio, chem, and computer science.

I got into residency at UCSF but the chief of surgery was a lesbian/rapist/molester so I dropped it and applied to masters programs in computer science.

I got into Columbia.

After a brief rendezvous with “quantum”, I decided to apply to medical school to get another MD. I got into Harvard/MIT MSTP.

Then the pandemic hit.

I did it remotely.

Graduated in 3 years.

Went on to do general surgery residency at Weill Cornell.

Then left.

Now where am I?

Doing a masters at NYU.

Also working on Wall Street, fulfilling a “re-venge arc” for my dad.

And I’m also going to reapply to medical school, hopefully for the “last” time.

Not nursing, definitely not. Not after the child molesters of Weill Cornell molested Davey.

And definitely, I won’t be joining a convent again.

Not on my mother’s deathbed either.

Peace.

†



07 — CREATIVE

When We Were Catholics

| Chapter 1 - Gloria

A flicker of storm crosses his face. His neck erect, his eyes scarlet yet ebony in unison, his Roman collar stretched tight across.

BY SHIRLEY XU · 30 JULY 2026 · 2 MIN READ

He hated it.
A flicker of storm crosses his face.
His neck erect, his eyes scarlet yet
ebony in unison, his Roman collar stretched
tight across.

He wonders what it will end.

Checking his watch, he looks for the time
when time is just a myth. In reality, there is

no time and there is no space.

We all exist in infinity.

Math is his strong suit. Ever since his days at
Oxford, where he did his thesis on fractals,
he's always excelled at using numbers to cal-
culate the sin of others.

These sinners, kneeling before him, their
hypocrisy unmatched by the corrupt politi-

cians that governed his city,

He preferred the Vatican. Italy, a country filled with ancient history and religion and culture, and none like it.

New York City is the bane of his existence.

He tugs at his Roman collar with his finger.

These sinners. I wish they were dead.

He remembered walking through Washington Square Park. A homeless woman sitting on a bench grabbed his rosary attached to his belt, yanking him to her. The scent of sin and smoke enveloped him, and he coughed.

Pulling his rosary away from her, he ran from the park, stumbling on the steps, back to his Church.

Away from the hard drugs, away from the unsolicited rapes, away from the prostitution, away from the thefts, away from the sinners.

These people don't know God.

Etched in his face, there existed a mark of contempt. Those who see him admire him. He was pastor, and he knew how to put on an act.

We all have affects, don't we, Fr. Edward?

A woman said this to him. Beautiful, a golden actress, glorious and tempting, magnetic and charming.

He caught the scent of sex before her many times, a perfume that aroused his very essence.

Holding his breath, he groans.

He wanted her. And she knew it.

Leaning back on his chair, he closes his eyes and sighs.

Another day. Another week.

Time doesn't stop, does it?

He remembered.

A child with a proper London accent. Blunt bangs, hair shorn to her head, more boy than girl, Chinese yet British, sitting on the steps of a Catholic church in London with a picnic basket filled to the brim with meat sandwiches, she looks up at him, smiling.

How can I be Catholic, Fr Edwards? Aren't I Asian?

He smiles bitterly to himself.

Passive aggression at its finest. From a toddler, nonetheless.

What is your name?

He picks up the sandwich from her basket besides her.

My name?

Sam.





08 — MIND

10 Financial Tips to Master Your Money

Personal finance advice is like Russian roulette. On one side, you have people telling you to invest your savings in a Roth IRA and buy S&P 500 index funds. On the other side, you have people...

BY SHIRLEY XU · 21 JULY 2026 · 5 MIN READ

Personal finance advice is like Russian roulette. On one side, you have people telling you to invest your savings in a Roth IRA and buy S&P 500 index funds. On the other side, you have people telling you to put your money in a high yield savings account and let it sit there gathering monthly

interest. Who should you believe? And what should you do?

I read an interesting book called *The Psychology of Money* that focuses more on the mental and emotional aspect of money than on complex financial strategies. How you grew up and what your perspective is on money defines your attitude and actions you

take with money. Not all of us were taught how to handle money from a young age, and financial wisdom becomes something we learn through making mistakes and trial and error.

Sometimes, we attach our emotions to money. Some of us are shopaholics, using shopping as a coping mechanism for deeper psychological issues. Some of us save every penny, refusing to buy even a coffee from a cafe when we are in need of a boost of energy when we're outside. Some of us measure life by coupons and discounts. What is the right way to live? What is the right way to think about money?

I think it's different for everyone. As it should be. Different people have different financial circumstances. Being realistic about how much you can spend versus how much you should be saving is a huge mental shift.

Pay Future You First

There's the 50-30-20 rule. You spend 50% of your income on necessities, 30% on wants, and 20% on savings. Automate the savings part. Once income hits your checking account, transfer at least 20% of it to a high yield savings account. This method protects the future you. Don't touch the savings account. Let the money in the savings account accumulate and draw interest.

Build a \$1000 starter buffer before anything else.

Gurus tell you to build an emergency fund of 3 to 6 months of living expenses, but building a \$15k savings fund when you're starting from 0 is daunting and potentially paralyzing.

Create a financial goal to build \$1000 in a HYSA (High-Yield Savings Account).

Once you've saved that amount, you'll feel a sense of relief. \$1000 is a lot of money. You can use it to pay off sudden medical expenses, last minute flights, high interest credit card debt, and food. Once you've exhausted the \$1000 fund, replenish it. And eventually, when you get more income, you can expand on it to build an emergency fund.

Get a job.

You should always be earning. If you don't have a job, get one. If you don't have a job that can be a career, get a part time or substitute job. and on the side, build on your skills and apply for jobs in your desired career sector.

Spending money without earning money is a sure way to get into debt fast. I've learned that the hard way. It costs to live, and not everyone has parents funding our spending habits. I've left those days behind in middle school.

Make money, or lose money. You should be earning more than you are spending.

No easy way around it.

Tackle Debt

We all obtain debt at some point in our lives. Student loans, high credit card payments, etc. If the debt has a high interest rate (anything above 7-8%), you need to pay it off. This debt will drag you down and slow your wealth building.

When paying off debt, finance experts argue between:

The Debt Snowball - Paying off the smallest balance first for a psychological win.

The Debt Avalanche - Paying off the highest-interest debt first to save the most money.

Focus on Big Wins Over Micro-Frugality

Cutting our daily coffee and skimping out on Netflix and HBO can save you money monthly, but at the cost of daily joy. You can cut your spending down to zero, at the cost of living like a hermit and increasing your stress levels.

Instead of stressing over tiny expenses, optimize the Big Three:

- Housing - get a roommate, move back in with your parents, renegotiate your lease, lower your rent. Lowering your rent by at least \$300/month can save you a huge amount of money
- Transportation - Spending money on Uber/Lift is a sure way to get bankrupt fast. Use public transit or drive a used car.
- Subscriptions - Audit your recurring subscriptions. If you don't use it, then lose it.

Give Every Dollar a Purpose

Money without a goal drains away unnoticed. To stay motivated, assign clear visual destinations to your funds.

Short Term Goals - vacation, emergency fund, the Macbook. Keep this cash safe in a HYSA where it earns interest without market risk.

Long Term Goals - retirement, long term wealth, FIRE (Financially Independent Retire Early). Put this money in a 401k or Roth IRA.

Track your money

Look over your monthly spreadsheets and track your expenses versus your income.

Maximize Income Leverage

Skills that are in high demand will give you an increased income. Build multiple income streams to diversify your income so you aren't reliant on a single employer.

Invest in Index Funds

S&P 500 index funds are a good way to accumulate wealth, they have an annual growth rate of 12-15%. Do your research before investing though. I'm not a trusted financial advisor, so take it with a grain of salt. Also, you might not be in a position where you can invest. Save your \$1000 before thinking of investing, and make sure your monthly income can support monthly investments.

Calculate your risk tolerance and money preferences.

We all have different personalities and preferences when it comes to money. Some of us like to trade and risk money on the stock market. Some of us like to put money away in a HYSA. Some of us like to spend money more than we earn money. Some of us like to be financially secure while some of us like to live life on the edge.

Sometimes, it matters to know what kind of person you are. Do you want to save money? Do you want to be financially secure? We all want to be wealthy, but people obtain this in different ways. Debating over the smartest way or the best way to obtain wealth is completely pointless and inefficient.

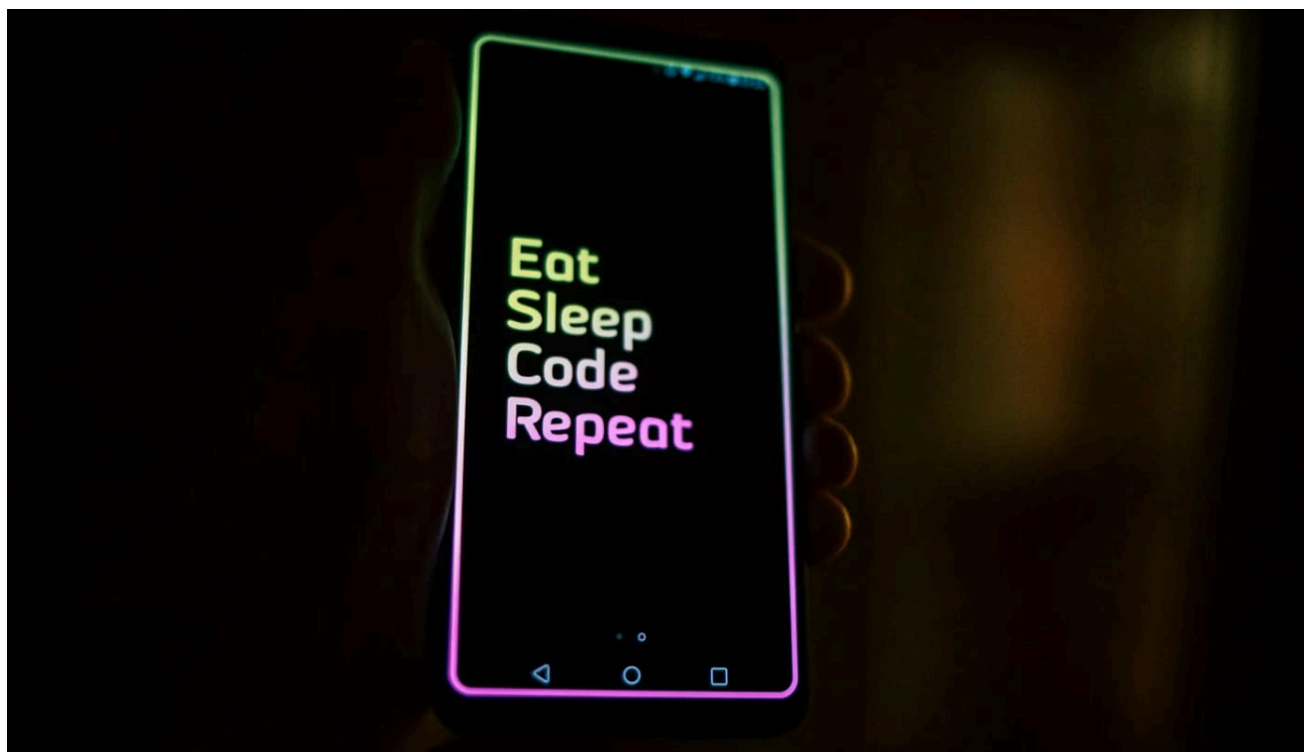
If you have the time to read this article, thank you. With the time you spent reading this,

you could have been earning money or building your skills to earn money. I hope this article was a good investment for you. If not, remind yourself that it was your choice.

We all have choices in life.

May the choices ever be in your favor.





09 — MIND

The Neuroscience Behind Programming: What Happens in the Brain When You Code?

It's 2am. I'm stuck in front of my laptop, squinting at microscopic code on my screen. Running on caffeine, I have to churn out this PR before its deadline that night. I could wait, I could go to...

BY SHIRLEY XU · 8 JULY 2026 · 5 MIN READ

It's 2am. I'm stuck in front of my laptop, squinting at microscopic code on my screen. Running on caffeine, I have to churn out this PR before its deadline that night. I could wait, I could go to sleep, but how can I? Even if there's more to life than

programming, exploring the beyond would be beyond the constraints of this assignment. I mumble a quick Hail Mary (have mercy on me, Lord) and blast some EDM to awaken my senses.

Programming is often seen as technical, a skill that many nerds have. You have to be logical, you have to be organized, you have to have Asperger's syndrome. Emotional intelligence isn't necessary for this realm of career, code isn't emotional, code isn't living, code isn't human. You have to learn appropriate syntax, algorithms, and software frameworks.

But there is something lying behind every line of code. As inhuman as it is, code needs the human brain to synthesize it, to invent it, to work it. A paradox, isn't it?

Whether you're building a website, debugging an application (cue meme of The Scream by Edvard Munch), designing a machine learning model, or solving a coding challenge, programming engages some of the most sophisticated cognitive systems human possess. It requires attention, memory, reasoning, creativity, language processing, and problem solving, all working together cohesively in real time.

While programming involves logic and analytical thinking, programmers often rely on brain networks associated with language comprehension, working memory, and pattern recognition.

When reading code, developers are not simply performing calculations. They are interpreting symbols, understanding relationships, predicting outcomes, and constructing mental models of how systems behave.

The Prefrontal Cortex: The Brain's Project Manager

One of the most important regions involved in programming is the prefrontal cortex. Located behind the forehead, the prefrontal cortex serves as the brain's executive control

center and is responsible for: planning, decision-making, attention regulation, goal management, problem-solving, and cognitive flexibility.

When designing software architecture or deciding between different algorithms, the prefrontal cortex evaluates possibilities, weighs trade-offs, and helps organize complex systems into manageable components. This region consumes significant mental energy, so prolonged coding sessions can lead to cognitive fatigue. That's why solving a difficult programming problem for several hours feels mentally exhausting, even though you're sitting at a desk.

Working Memory: The Brain's Temporary Workspace

Working memory is the brain's ability to temporarily store and manipulate information while completing a task. While coding, you need to remember function parameters, variable names, API responses, database schemas, component relationships, business requirements, error messages, and logic across multiple files.

Working memory has limited capacity, so when too much information must be tracked simultaneously, cognitive overload occurs, increasing likelihood of bugs and slowing problems solving.

Experienced developers reduce this burden by breaking large problems into smaller modules, writing clearer documentation, using descriptive variable names, and relying on diagrams or notes to offload information from memory.

Pattern Recognition

As you gain experience in programming, your brain stores thousands of programming “chunks.” Instead of seeing individual lines of code, experienced developers recognize familiar structures almost instantly.

Authentication flows, CRUD operations, API routes, design patterns, database queries, error-handling structures, common algorithms. This process is known as chunking.

Chunking reduces cognitive load by allowing the brain to treat multiple related pieces of information as a single unit. It frees up mental resources for solving higher-level problems instead of focusing on syntax.

Neuroplasticity

The brain adapts through neuroplasticity. Every time you solve a problem, neural pathways associated with reasoning, memory, and abstraction become stronger. Repeated practice strengthens these connections through long-term potentiation (LTP). As these pathways become more efficient, tasks that once required intense concentration gradually become automatic. Concepts that seemed impossible like recursion, asynchronous programming, or object oriented design feel intuitive after sufficient practice.

Learning to code is physically reshaping the neural architecture of your brain.

Debugging

Debugging is one of the most cognitively demanding aspects of programming. Your brain has to — form hypotheses, test assumptions, eliminate incorrect possibilities, update mental models, search for inconsistencies and maintain attention over long periods of time.

Attention

Programming demands sustained attention. Unfortunately, modern workplaces are filled with interruptions. Slack notifications, emails, meetings, texts, social media, context switching between projects. Each interruption forces the brain to reorient itself, a process known as task switching. The brain rapidly switches attention, and each switch incurs a mental cost.

Frequent disruptions reduce productivity, increase errors, and make complex reasoning more difficult. After an interruption, it takes several minutes to regain former concentration.

Creating uninterrupted periods of deep work, such as 60-90 minute coding sessions, allows the brain to fully engage with challenging problems.

Sleep

During sleep, the brain — consolidates newly learned programming concepts, strengthens neural connections, integrates knowledge across projects, enhances creative problem solving, and removes metabolic waste products that are toxic to your brain.

Sleep often reorganizes information, allowing your brain to approach problems with greater clarity.

Getting 7-9 hours of quality sleep improves learning, attention, concentration, memory, and decision making.

Exercise

Physical activity can make your brain a better programmer. Regular exercise increases cere-

bral blood flow and stimulates the release of brain-derived neurotrophic factor (BDNF), a protein that supports:

- Learning
- Memory
- Neuroplasticity
- Attention
- Executive function

Even a 20 minute walk before beginning work can improve concentration and reduce mental fatigue.

Using AI to Reduce Cognitive Load

Modern AI tools have become valuable coding assistants because they reduce unnecessary cognitive load. They can help by:

- Explaining unfamiliar codebases
- Generating boilerplate code
- Suggesting documentation
- Creating unit tests
- Identifying potential bugs
- Translating between programming languages
- Summarizing APIs
- Providing examples of common design patterns

By automating repetitive tasks, AI allows developers to devote more mental energy to software architecture, critical thinking, debugging, and solving novel problems.

The most effective programmers use AI as a collaborator while developing a deep understanding of the systems they build.

How to Optimize Your Brain For Coding

- Practice consistently. Short, regular coding sessions strengthens neural pathways more effectively than infrequent marathons
- Build projects. Applying concepts in real applications creates stronger and more flexible memories than passive tutorials
- Break problems into modules. Smaller tasks reduce working memory demands and makes debugging easier.
- Review old code. Revisiting previous projects reinforces learning and highlights growth.
- Protect deep focus time. Minimize interruptions during challenging work.
- Exercise regularly. Physical activity supports learning and cognitive flexibility.
- Prioritize sleep. Memory consolidation and insight depend on adequate rest.
- Use AI thoughtfully.

Final Thoughts

Programming is much more than typing syntax into a text editor. It is a sophisticated cognitive activity that engages memory, attention, language, reasoning, creativity, and problem-solving. Every bug you fix, algorithm you optimize, and application you build strengthens neural pathways through neuroplasticity, making future challenges easier to tackle.

By understanding the neuroscience behind coding, developers can move beyond simply working harder. Building consistent practice habits, protecting deep-focus time, managing cognitive load, exercising regularly, sleeping well, and leveraging AI wisely all help the brain perform at its best.

Great programmers are not born with fundamentally different brains. They become great by repeatedly training their brains to recognize patterns, solve problems, and think systematically. Every project is another opportunity to refine those neural circuits, expanding

not only your technical skills but also your capacity to learn, adapt, and create.

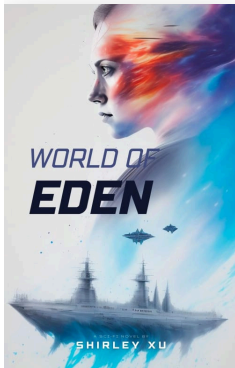
"Do you see a man skillful in his work? He will stand before kings; he will not stand before obscure men." - Proverbs 22:29

†

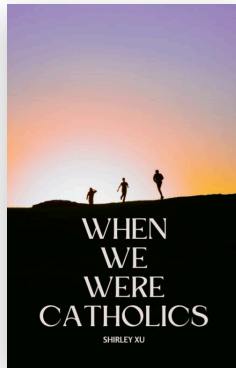
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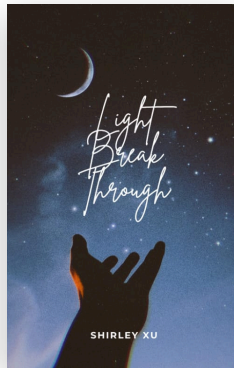
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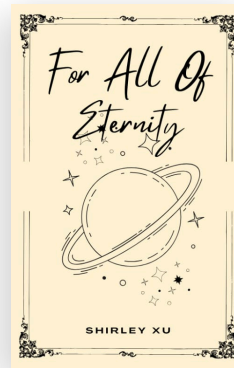
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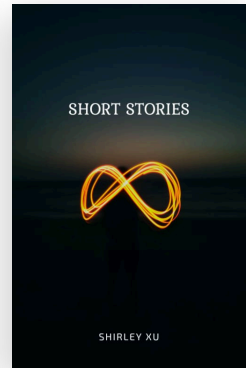
When We Were Catholics



Light Break Through



For All Of Eternity



Short Stories

MIND & SPIRIT

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Ad Majorem Dei Gloriam